

Problem 3

Problem 1 The graph of a function f is given below. Use this graph to answer the following questions.

alt=graph of piecewise-defined function for finding domain, range, and limit values from graph.

(a) Find the domain of f .

Explanation. The domain of f is $(-\infty, -2) \cup (-2, \infty)$.

(b) Find the range of f .

Explanation. The range of f is $(-\infty, \infty)$.

(c) Find the following values.

(i) $\lim_{x \rightarrow -2} f(x) =$

Explanation. $\lim_{x \rightarrow -2} f(x) = 0$

(ii) $f(-2) =$

Explanation. $f(-2)$ is undefined

(iii) $f(-5) =$

Explanation. $f(-5) = -2$

(iv) $\lim_{x \rightarrow 0^+} f(x) =$

Explanation. $\lim_{x \rightarrow 0^+} f(x) = 0$

(v) $\lim_{x \rightarrow 0} f(x) =$

Explanation. $\lim_{x \rightarrow 0} f(x)$ is undefined